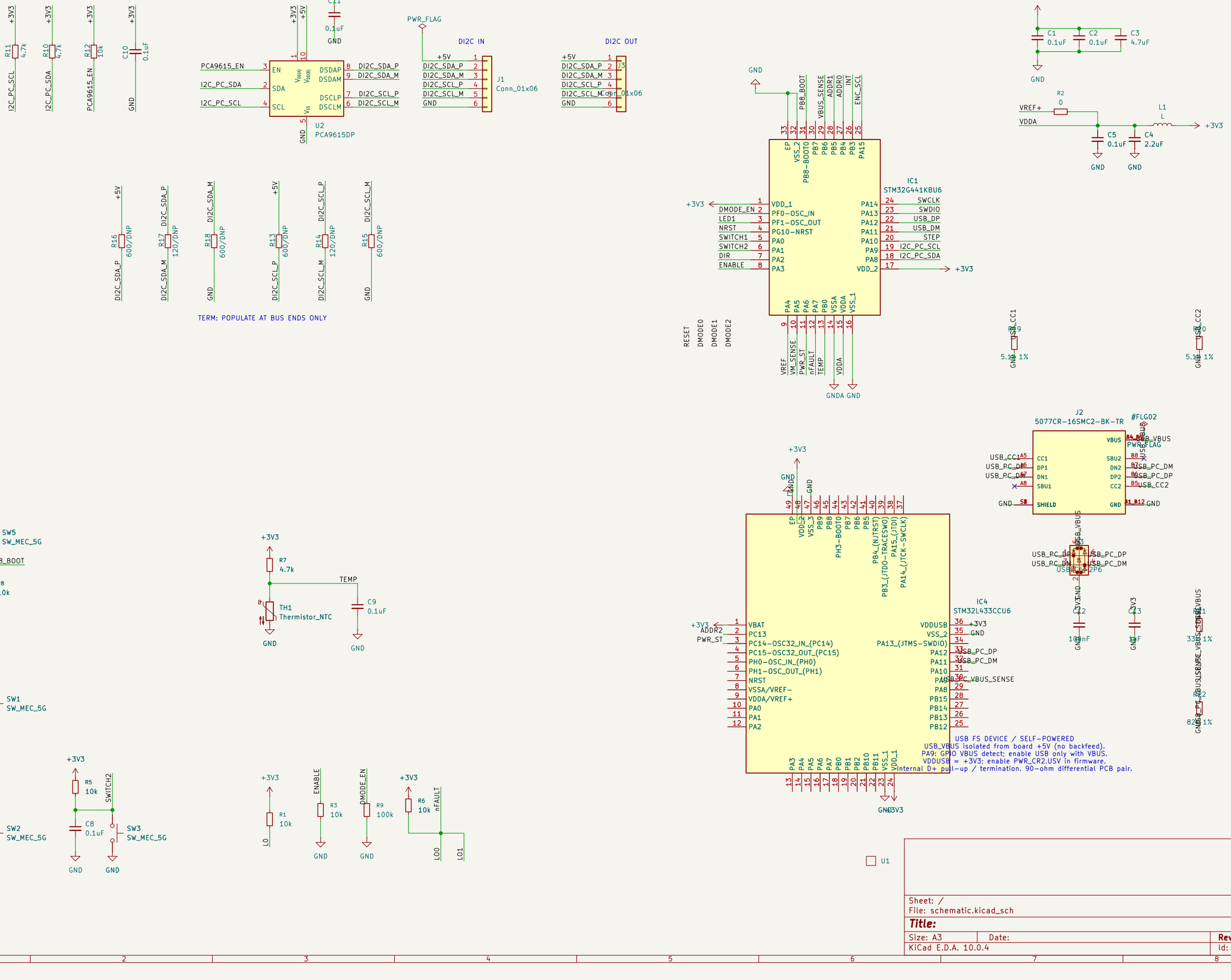

R11 4.7k
R10 4.7k
R12 10k
C10 0.1uF
I2C_PC_SCL
I2C_PC_SDA
PCA9615_EN
GND
+3V3
+5V
C11 0.1uF
U2
PCA9615DP
EN
SDA
SCL
GND
DSDAP
DSDAM
DSCLP
DSCLM
DI2C_SDA_P
DI2C_SDA_M
DI2C_SCL_P
DI2C_SCL_M
PWR_FLAG
DI2C IN
J1
Conn_01x06
DI2C OUT
J3
R16 600/DNP
R17 120/DNP
R18 600/DNP
R13 600/DNP
R14 120/DNP
R15 600/DNP
TERM: POPULATE AT BUS ENDS ONLY
GND
IC1
STM32G441KBU6
VDD_1
PF0-OSC_IN
PF1-OSC_OUT
PG10-NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB10
PB11
VSS_1
VDD_2
PA14
PA13
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
SWCLK
SWDIO
USB_DP
USB_DM
STEP
I2C_PC_SCL
I2C_PC_SDA
DMODE_EN
LED1
NRST
SWITCH1
SWITCH2
DIR
ENABLE
RESET
DMODE0
DMODE1
DMODE2
VREF
VM_SENSE
PWR_ST
nFAULT
TEMP
VDDA
VSSA
GNDA
PB8_BOOT
VBUS_SENSE
ADDR1
ADDR0
INT
ENC_SCL
C1 0.1uF
C2 0.1uF
C3 4.7uF
R2 0
L1 L
C5 0.1uF
C4 2.2uF
VREF+
VDDA
USB_CC1
USB_CC2
5.1k 1%
J2
5077CR-16SMC2-BK-TR
#FLG02
VBUS
SBU2
DN2
DP2
CC2
SBU1
SHIELD
USB_VBUS
USB_PC_DM
USB_PC_DP
PWR_FLAG
USB2P6
C12 100nF
C13 1uF
R21 33
R22 82
1%
C14 100nF
SW5 SW_MEC_5G
PB8_BOOT
R8 10k
SW1 SW_MEC_5G
C6 0.1uF
SW2 SW_MEC_5G
C7 0.1uF
SW3 SW_MEC_5G
C8 0.1uF
R4 10k
R5 10k
R7 4.7k
TH1 Thermistor_NTC
C9 0.1uF
TEMP
R1 10k
R3 10k
R9 100k
R6 10k
LO0
LO1
U1
IC4
STM32L433CCU6
VBAT
PC13
PC14-OSC32_IN_(PC14)
PC15-OSC32_OUT_(PC15)
PH0-OSC_IN_(PH0)
PH1-OSC_OUT_(PH1)
VSSA/VREF-
VDDA/VREF+
PA0
PA1
PA2
VDDUSB
VSS_2
PA13_(JTMS-SWDIO)
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
USB_VBUS_SENSE
USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal D+ pull-up / termination. 90-ohm differential PCB pair.
Sheet: /
File: schematic.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1